

PROJECT PLANNING PHASE

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Date	2 August 2026
Team ID	Not Specified
Project Name	Digitize Metro Ticket Booking using ServiceNow
Maximum Marks	5 Marks

Project Overview

Digitize Metro Ticket Booking using ServiceNow to enable fast, accessible, and automated ticket generation through QR codes. The system allows users to book tickets by selecting source and destination stations and entering travel details. It automatically calculates the applicable fare and generates a digital ticket with a QR code. The solution uses Service Catalog for ticket booking requests, Flow Designer for workflow automation and fare/ticket processing, and Notifications for booking confirmations and ticket delivery. The objective is to streamline the metro ticket booking process and provide users with a convenient, efficient, and seamless travel experience.

Problem Statement

Traditional metro ticket booking can involve queues, manual processing, and delays during peak travel hours. Users may also need to spend additional time selecting stations, calculating fares, and obtaining physical tickets. The proposed ServiceNow-based solution addresses these challenges by digitizing the ticket booking process. Users can submit a booking request through a Service Catalog item, select the required travel details, receive an automatically calculated fare, and obtain a digital QR-code ticket through an automated workflow and notification. This improves accessibility, reduces manual effort, and supports a faster and more efficient ticketing experience.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The following product backlog defines the major functional requirements (epics), user stories, tasks, priorities, and estimated story points for the metro ticket booking solution.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Service Catalog Booking	USN-1	As a user, I can access a metro ticket booking catalog item and submit a ticket booking request.	3	High	
Sprint-1	Station & Travel Details	USN-2	As a user, I can select the source station, destination station, travel date, and number of passengers.	3	High	
Sprint-1	Fare Calculation	USN-3	As a user, I can view the automatically calculated fare based on my selected stations and travel details.	3	High	
Sprint-2	Automated Ticket Generation	USN-4	As a user, I can receive a digital metro ticket generated automatically after a successful booking request.	3	High	
Sprint-2	QR Code Ticket	USN-5	As a user, I can receive a unique QR code with my digital ticket for convenient ticket verification.	3	High	
Sprint-2	Flow Automation	USN-6	As a system administrator, I can use Flow Designer to automate booking validation, fare processing, ticket generation, and status updates.	5	High	

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Notifications	USN-7	As a user, I receive a notification confirming my booking and providing access to the generated digital ticket.	2	Medium	_____
Sprint-3	Booking Status	USN-8	As a user, I can view the status of my metro ticket booking request.	2	Medium	_____
Sprint-3	Testing & Validation	USN-9	As a project team member, I can test the complete booking workflow and validate QR ticket generation and notifications.	3	High	_____

Sprint Schedule

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	12	6 Days	27 Jul 2026	01 Aug 2026	_____	_____
Sprint-2	11	6 Days	03 Aug 2026	08 Aug 2026	_____	_____
Sprint-3	10	6 Days	10 Aug 2026	15 Aug 2026	_____	_____

Project Tracker, Velocity & Burndown Chart: (4 Marks) Project

Tracker

Work Item	Sprint	Owner	Priority	Status	Story Points	Target Date
Service Catalog booking form	Sprint-1	_____	High	Not Started	3	01 Aug 2026
Station and travel details	Sprint-1	_____	High	Not Started	3	01 Aug 2026
Automated fare calculation	Sprint-1	_____	High	Not Started	3	01 Aug 2026
Digital ticket generation	Sprint-2	_____	High	Not Started	3	08 Aug 2026
QR code generation	Sprint-2	_____	High	Not Started	3	08 Aug 2026
Flow Designer automation	Sprint-2	_____	High	Not Started	5	08 Aug 2026
Booking confirmation notifications	Sprint-3	_____	Medium	Not Started	2	15 Aug 2026
Booking status tracking	Sprint-3	_____	Medium	Not Started	2	15 Aug 2026
End-to-end testing and validation	Sprint-3	_____	High	Not Started	3	15 Aug 2026

Velocity

For this project, the planned total work is 33 story points across three sprints. Assuming a 6-day sprint duration and an estimated team capacity of 20 story points per sprint, the average velocity per iteration unit (story points per day) is calculated as follows:

Average Velocity (AV) = Team Velocity / Sprint Duration = 20 / 6 = 3.33 story points per day.

The planned backlog is distributed across three sprints according to priority and dependencies. Actual velocity should be updated after each sprint based on the story points completed.

Burndown Chart

A burndown chart is a graphical representation of the amount of work remaining versus time. For this project, the chart can be maintained throughout the sprints by tracking the remaining story points after each working day. The ideal burndown begins with the total planned story points and decreases toward zero as user stories are completed. The actual remaining work should be updated based on the team's completed story points.

Sprint / Day	Planned Remaining Story Points	Actual Remaining Story Points
Start	33	33
Day 1	29.67	_____
Day 2	26.33	_____
Day 3	23.00	_____
Day 4	19.67	_____
Day 5	16.33	_____
Day 6	13.00	_____
Sprint-1 Completion	21	_____
Sprint-2 Completion	10	_____
Sprint-3 Completion	0	_____

Project Outcome

The planned ServiceNow solution will provide a structured and automated metro ticket booking workflow. By combining Service Catalog, Flow Designer, and Notifications with digital QR-code ticket generation, the system is designed to reduce manual processing, improve booking accessibility, and provide users with faster digital ticket delivery. The completed project will demonstrate an end-to-end automated ticket booking process suitable for a ServiceNow implementation.